

		Option D: The equation does not adhere to $\frac{1}{g} \frac{d^2x}{dt^2}$.
13	D	Frictional force by P on Q is the restoring force for Q. Without friction, Q will not move as P slides underneath it. Net force on Q is provided for by the frictional force by P on Q. friction = $ma = m\omega^2 x$ $5.0 = 0.2 \{(2\pi)(1.5)\}^2 A$ $A = 0.28 \text{ m}$
14	B	The components of the particle's motion in the horizontal x -direction is simple harmonic. Hence